MATH317 · NUMERICAL METHODS

Final Drill Lectures 2–6

كل أمثلة السلايدات في ملف واحد. حلها بنفسك، وبعدها قارن بملف الحل

المحاضرة	الأسئلة
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Q1 Significant digits L2 · 1.1

How many significant digits does each number have?

a) 0.0035 b) 30.16 c) 12.345

Q2 Absolute and relative error L2 · 1.1

a) x = 0.00347 is rounded to 0.0035.

b) y = 30.158 is rounded to 30.16.

Find the absolute error and the relative error in each case. Which one is more accurate?

L2 · 1.1 **Rounding and chopping** Q3

Write each number to two digits, once by rounding and once by chopping:

0.217 0.365 0.475 0.592

L2 · 1.1 **Horner's algorithm** Q4

Use Horner's algorithm to find $P(3)$ for $P(x) = 4x^3 - 8x^2 + 14x - 32$.

L3 · 1.2 **Using a known series** Q5

a) Approximate $\ln(1.1)$ using five terms of the series for $\ln(1 + x)$.

b) Approximate e^8 using six terms of the series for e^x. The true value is 2980.96. What do you notice?

L3 · 1.2 **Taylor series of a polynomial (derivatives)** Q6

Find the Taylor series of $f(x) = x^3 - 2x^2 + 4x - 1$ about $c = 2$.

L3 · 1.2 **Complete Horner** Q7

Using the complete Horner's algorithm, find the Taylor expansion of $p(x) = x^4 - 4x^3 + 7x^2 - 5x + 2$ about $r = 3$.

L3 · 1.2 **Taylor in powers of h** Q8

Expand $\sqrt{(1 + h)}$ in powers of h up to h^2, then approximate $\sqrt{1.00001}$.

L3 · 1.2 **Alternating series** Q9

How many terms of the series for sin 1 are needed so that the error is less than $\frac{1}{2} \times 10^{-6}$?

L3 · 1.2 **Examples 5 and 6 (one step each)** Q10

a) Write the Taylor series of e^x at $c = 0$ with its remainder after n terms.

b) For $f(x) = \ln(1 + x)$, find $f'''(0)$.

c) For which x does the Taylor series of $\ln(1 + x)$ at $c = 0$ represent $\ln(1 + x)$ (as shown in the slides)?

L4 · 2 **Single precision** Q11

Represent -52.234375 in single precision. Give the answer in hexadecimal.

L4 · 2 **Double precision** Q12

Represent the same number, -52.234375, in double precision.

L4 · 2 **Hex to decimal** Q13

Which decimal number is stored in single precision as $[45DE4000]_{16}$?

L4 · 2 **Roundoff error** Q14

A machine keeps 5 decimal digits. Add $x = 0.37218 \times 10^{4}$ and $y = 0.71422 \times 10^{-1}$. Find fl(x + y) and the relative error.

L5 · 3.1 **Bisection** Q15

$f(x) = x^3 - 3x + 1$ on $[0, 1]$.

a) Show that f has a root in $[0, 1]$.

b) Apply the bisection method for $n = 0, 1, 2$. Give c, f(c) and the error in a table.

L5 · 3.1 **Number of steps** Q16

a) Write the bound on $|r - c_n|$ after n bisection steps.

b) How many steps give full single precision when a = 16 and b = 17?

L5 · 3.1 **False position** Q17

$f(x) = x^3 - 2x - 3$ on [0, 2]. Apply two steps of the false position method.

L6 · 3.2 **Newton's method** Q18

If $f(x) = x^3 - x + 1$ and $x_0 = 1$, what are x_1 and x_2 in the Newton iteration?

L6 · 3.2 **Newton's speed** Q19

Solve $x^3 + x = 2x^2 + 3$ with Newton's method starting from $x_0 = 3$. Find x_1 and x_2.

L2 – L6 **True / False** Q20

a) A result is never more accurate than the least accurate number used.

b) The relative error is a better measure of accuracy than the absolute error.

c) A Taylor series converges rapidly far from the expansion point.

d) The Mean-Value Theorem is a special case of Taylor's theorem with n = 0.

e) Single precision uses an 8-bit exponent with a bias of 127.

f) In single precision the leading 1 is stored in the fraction field f.

g) The bisection error after n steps is at most $(b - a) \div 2^n$.

h) If $f(a) \cdot f(b) > 0$, then f has no root in [a, b].

i) Newton's method always converges.

j) Newton's method needs the derivative $f'(x)$.